



2" 45° E/low

CERTIFIED MATERIAL TEST REPORT
CERTIFICATE OF COMPLIANCE
CERTIFICATE # 177766

Tube Forgings of America
5200 N.W. Front Avenue
Portland, OR 97210
ISO 9001:2015 Certified
5/26/2022

Sold To	DNOW LP #ND16 357 Central Ave North Dunn Center, ND 58626-2610	Ship To	DNOW LP #ND16 357 Central Ave North Dunn Center, ND 58626-2610
Customer ID	34326		
TFA Sales Order	3182051-1		
Purchase Order	26417656		

TFA Part ID	05055	Heat Code	A2402
Description	2 XH LR 45	Grade	WPB
Quantity	4	Raw Material Heat	J2L1024
Customer Part ID	3690	Process	WPB-H1
		Mill	NIPPON STEEL

Austenitized @: Held For:	mins/min	Tempered @: Held For:	mins.	Stress Relieved @: Held For:	mins.	Medium:
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Heat Analysis Chemistry

C	Mn	P	S	Si	Mo	Cr	Ni	Cu	V	Co	B	Ti	Al	N	Co	Ca	C
0.18	1.09	0.021	0.003	0.25	0.01	0.06	0.02	0.02	0.00								

Physical Results

ID	Type	Yield	Tensile	% Elong	R/A %	Gage Length	HBW	In-Process Test 1	In-Process Test 2	Tensile Orientation	Test Bar Type
1	PSI	50,000	75,900	55.00		2"		138	143	LONGITUDINAL	FULL SECTION

Charpy Test Results

Impacts				Lateral Expansion (Mils)			% Shear				V notch test at degrees F	Coupon Size	Original
1	2	3	Avg	1	2	3	1	2	3	Avg			

Notes/Comments

1. THESE FITTINGS WERE HOT FORMED BETWEEN 1160F AND 1800F AND AIR COOLED. 2. THE HARDNESS DOES NOT EXCEED 210 BHN PART TO PART. 3. STARTING MATERIAL IS SEAMLESS PIPE WHICH MEETS THE MELTING, CHEMICAL AND PHYSICAL REQUIREMENTS OF ASTM/ASME A/SA 106 GR B. 4. THIS PRODUCT IS SEAMLESS.

5. MATERIAL IS MANUFACTURED TO ASTM A234-19 GR WPB, ALSO PER ASME SEC II SA234 GR WPB-2019 EDITION. 6. DIMENSIONS TO ASME B16.9-2018 3. PER NACE MR0175/ISO 15156-2-2015 EDITION, AND NACE MR-0103 2015 EDITION. 7. MANUFACTURED IN AN ISO 9001 REGISTERED FACILITY, CERTIFICATE # 30248. 8. TFA, INC. ORGANIZATIONAL STRUCTURE CERTIFICATION COMPLIES WITH EN 10204-2004 TYPE 3.1 AND 3.2. 9. TFA INC. FITTINGS SATISFY THE MATERIAL REQUIREMENTS OF (PED) 2014/68/EU ANNEX I, PARA 4.3 7. FITTINGS ARE MANUFACTURED IN THE USA. RAW MATERIAL IS FROM DOMESTIC GLOBAL SOURCES. 8. FITTINGS WERE MANUFACTURED, SAMPLED, TESTED AND INSPECTED IN ACCORDANCE WITH REQUIREMENTS AND SPECIFICATIONS AND WERE FOUND TO MEET THE REQUIREMENTS.

Grade: WPB

Scot Marshall, Manager Quality

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